

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q22-2114

Dighton Power Plant 100 kV  
May 30, 2023

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2114. The Interconnection Customer proposes a 69.7 MW solar photovoltaic facility interconnecting at the Dighton Power Plant 100 kV substation in the Ohio area. The request was received on June 14, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 727 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,946,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                              | Contingency                                   | Loading (%) | Rating (MVA) | DFAX (%) |
|-------------------------------------------------|-----------------------------------------------|-------------|--------------|----------|
| Barret – Roseau County 100 kV                   | Harper – Monte Del Estado 6303 100 kV         | 157.9       | 234          | 7.0      |
| Roseau County – S Whitehall 100 kV              | Monte Del Estado 6303 – Harper 100 kV line    | 136.2       | 172          | 7.1      |
| Hassen Cassel – Drlington Rd 100 kV             | Darlington Road – Wayne Tace 100 kV L         | 130.1       | 189          | 5.1      |
| Hassen Cassel – Muldoon Mill 100 kV             | Darlington Rd – Wayne Trace Substation 100 kV | 127.3       | 204          | 5.1      |
| Pt A – Lake Marion 100 kV L                     | State St – Robison Park 230 kV                | 118.2       | 224          | 5.0      |
| Brret – Pt A 100 kV                             | State St – Robison Park 230 kV Line           | 115.9       | 247          | 5.0      |
| Harper – Monte Del Estado 6303 100 kV           | Barret – Roseau County 100 kV                 | 103.6       | 423          | 5.8      |
| Hillcrest Substation – Muldoon Mill 100 kV line | Wayne Trace – Darlington Road 100 kV          | 102.4       | 435          | 8.1      |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                        | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Roseau County – Barret Substation 100 kV        | Rebuild line with higher-capacity conductor | \$7,327,000   | 23% (\$1,653,000)     |
| Roseau County – S Whitehall 100 kV              | Rebuild line with higher-capacity conductor | \$8,212,000   | 21% (\$1,708,000)     |
| Hassen Cassel – Darlington Rd 100 kV line       | Rebuild line with higher-capacity conductor | \$2,206,000   | 18% (\$402,000)       |
| Hassen Cassel – Muldoon Mill Substation 100 kV  | Rebuild line with higher-capacity conductor | \$1,863,000   | 18% (\$339,000)       |
| Pt A – Lake Marion 100 kV L                     | Rebuild line with higher-capacity conductor | \$2,612,000   | 35% (\$906,000)       |
| Barret – Pt A 100 kV                            | Rebuild line with higher-capacity conductor | \$13,960,000  | 35% (\$4,841,000)     |
| Harper – Monte Del Estado 6303 100 kV           | Reconductor line                            | \$2,597,000   | 2% (\$64,000)         |
| Hillcrest Substation – Muldoon Mill 100 kV line | Reconductor line                            | \$1,263,000   | 3% (\$33,000)         |
|                                                 | Total allocated to project                  |               | \$9,946,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.